

VAPOROSAS



Estudio de consumos

VAPOROSAS PROJECT

Spanish Steam Locomotives Consumption Study

Executive Report

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1. EXECUTIVE SUMMARY

Objective: Compare water and coal consumption of four Spanish steam locomotives (MZA 1701, MZA 1801, AND240-4253, RN141-2106 "Mikado") through Open Rails simulation with manual driving and realistic tolerances.

Methodology: 73 consumption records across three routes (long distance 152 km, commuter 53 km, ramp 8 km). Statistical analysis in R (descriptives, regressions, comparisons).

Main conclusion: RN141-2106 (Mikado) is the most balanced locomotive, while MZA 1701 is the most economical for long distance. AND240-4253 stands out for predictability and MZA 1801 for speed, despite high coal consumption.

Ranking	Locomotive	Key takeaway
1st	RN141-2106 (Mikado)	Most balanced. Excellent water (133.9 L/km) and coal (12.6 kg/km)
2nd	MZA 1701	Most economical for long distance. 2.93 €/km. Excellent coal (12.3 kg/km)
3rd	AND240-4253	Most predictable (CV 34-47%). Ideal for short-distance services
4th	MZA 1801	Fastest (47.8 km/h), but consumes ~3x more coal (38.7 kg/km)

2. LOCOMOTIVES ANALYZED

Name	Configuration	Adhesive weight (t)	Power (HP)	Era
MZA 1701	2-4-1	64.3	2231	Transition (1915)
MZA 1801	2-4-1	77.0	2434	Steam maturity (1939)
AND240-4253	2-4-0	77.6	2549	Raw power (1925)
RN141-2106	1-4-1	72.0	2000	Steam maturity (1945)

Table 1. Locomotives analyzed

3. TEST ROUTES

Route	Length (km)	Type of service	Characteristics
Vallarizatoria (Soria - Burgos)	152.26	Long distance	Express service
Eje Central Andaluces (Granada – Loja)	52.74	Regional service	Stopping service
CTN Test Route	8.30	Experimental service	Pure ramp

Table 2: Test routes

4. KEY RESULTS

4.1 Average consumption (all tests)

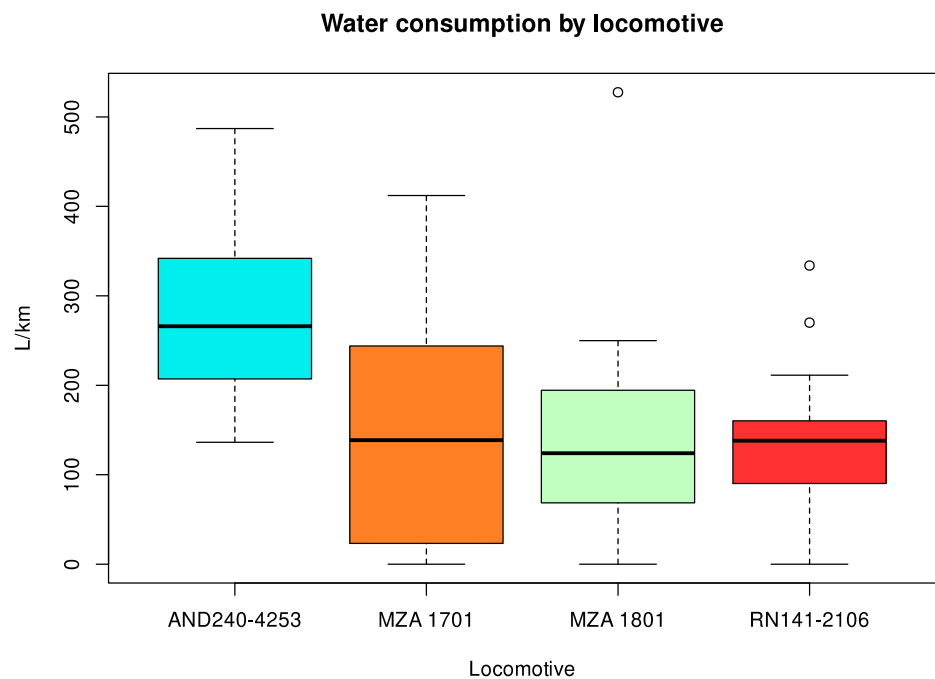


Figure 1: Water consumption Boxplot.

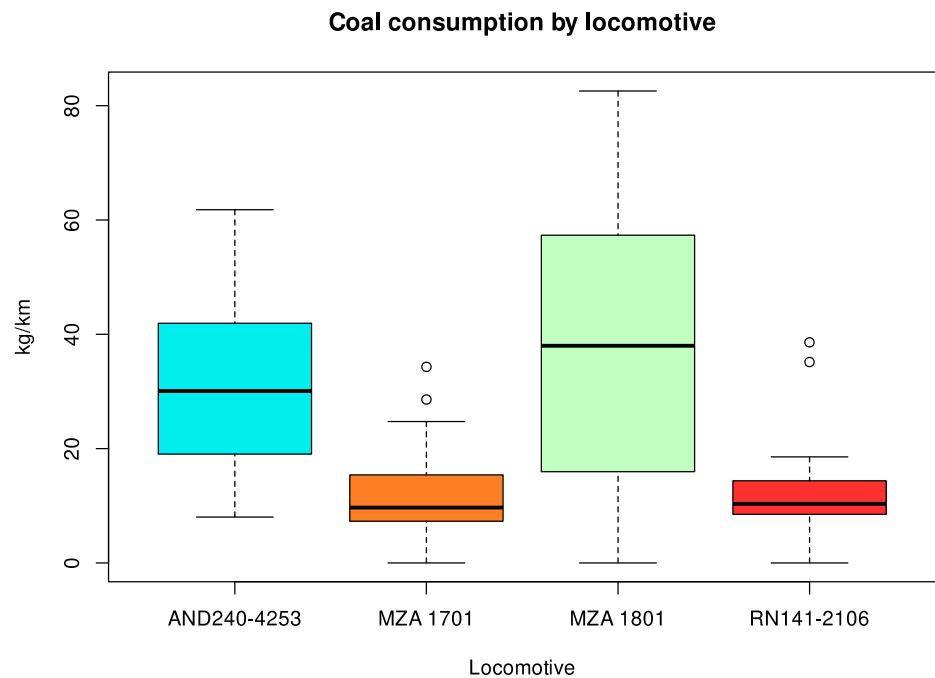


Figure 2: Coal consumption boxplot

Locomotive	Water(L/km)	Coal (kg/km)	Speed (km/h)
RN141-2106	133.9	12.6	45.2
MZA 1701	149.6	12.3	45.1
MZA 1801	144.5	38.7	47.8
AND240-4253	277.8	32.2	43.5

Table 3: Average consumptions

The Mikado is best for water, MZA 1701 for coal, MZA 1801 for speed.

4.2 By terrain type (Vallarizatoria)

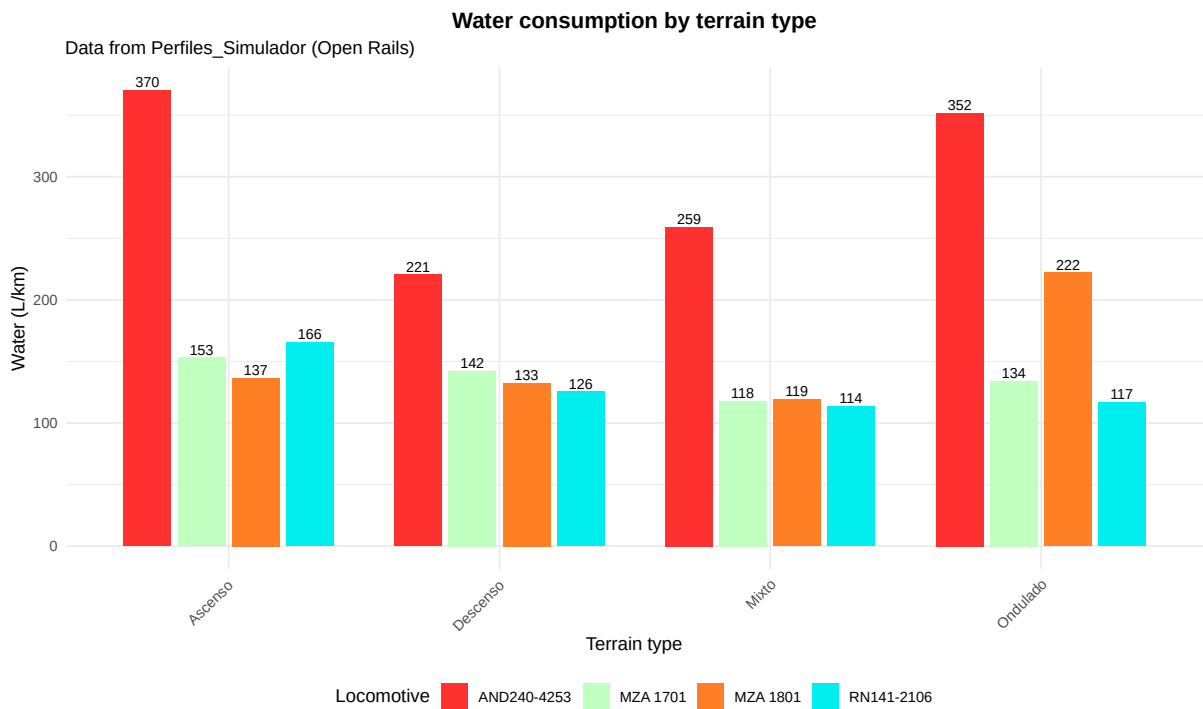


Figure 3: Water consumption by terrain

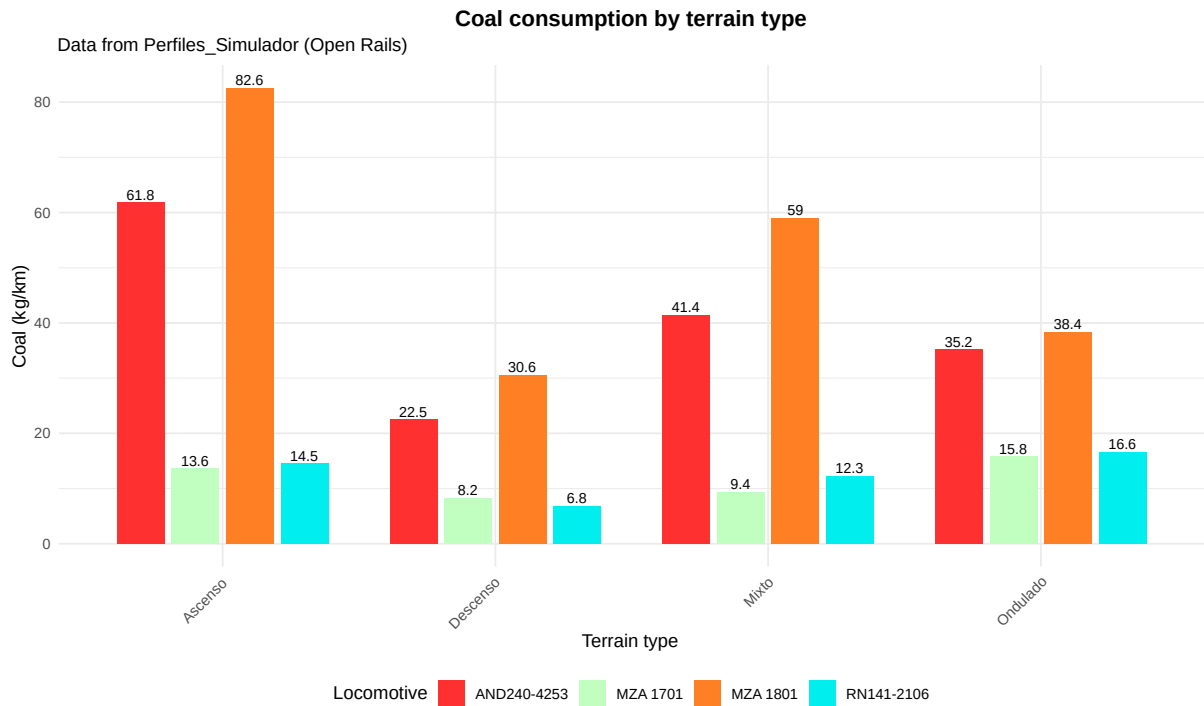


Figure 4: Coal consumption by terrain

Zone	Best locomotive (coal)	Runner-up
Climb (uphill)	MZA 1701	RN141-2106
Descent (downhill)	RN141-2106	MZA 1701
Rolling	MZA 1701	RN141-2106

Table 4: Best locomotive by zone

MZA 1701 is the king of hills.

4.3 Economic analysis

Prices: Water 0.0036 €/L | Coal 0.24 €/kg

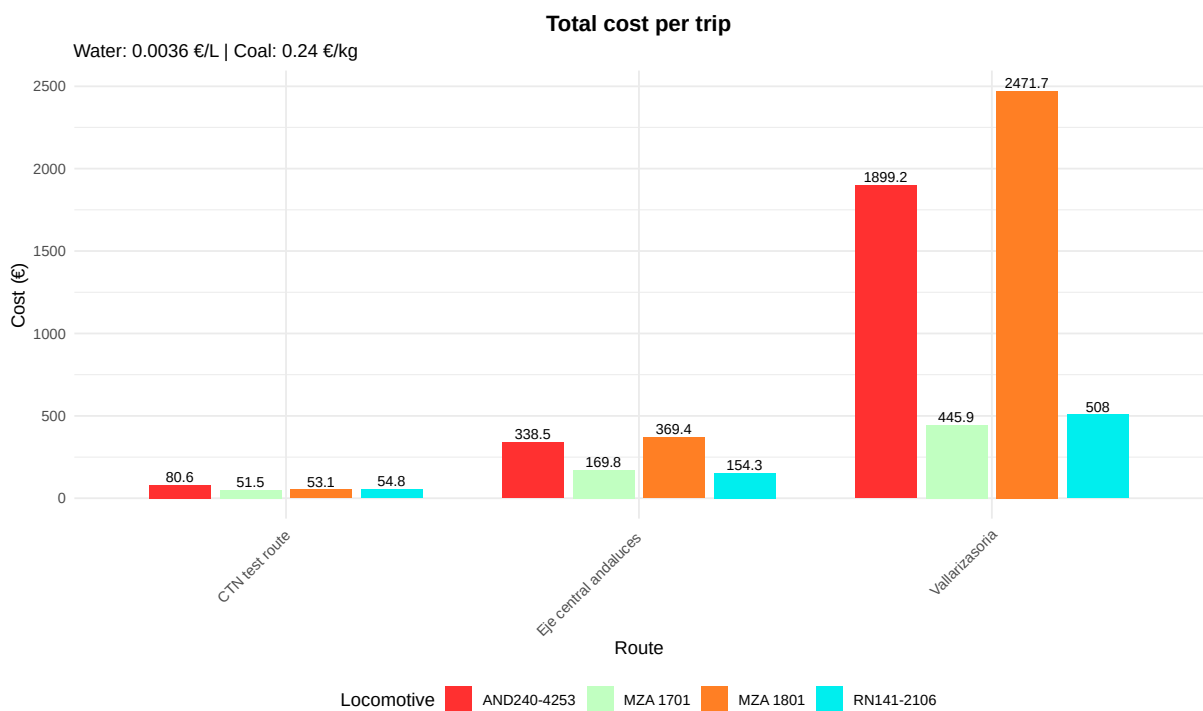


Figure 5: Cost per trip

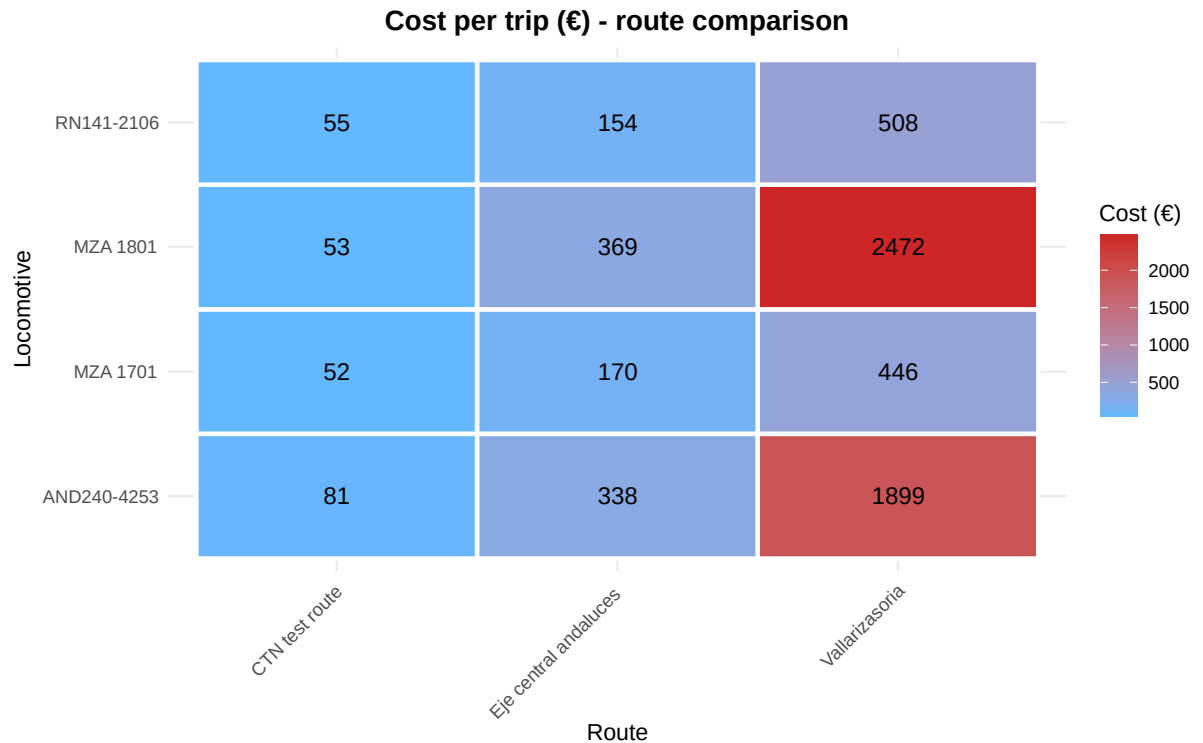


Figure 6: Cost heatmap

Locomotive	Vallarizatoria (152 km)	Eje Central (53 km)	Test Route (8 km)
MZA 1701	445.87	169.79	51.49
RN141-2106	507.95	154.35	54.74
AND240-4253	1899.17	338.50	80.56
MZA 1801	2471.69	369.39	53.07

Table 5: Total cost per trip (€)

MZA 1701 dominates long distance, Mikado dominates commuter services.

4.4 Predictability (Coefficient of Variation)

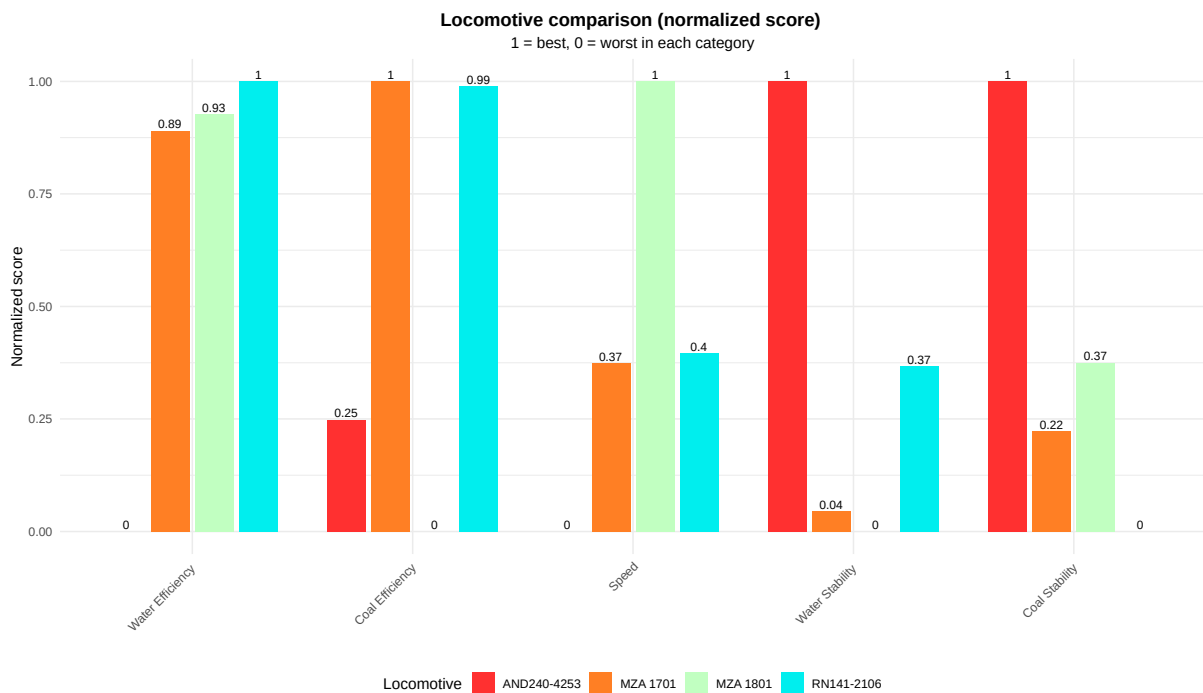


Figure 7: Comparative radar chart

Locomotive	Water CV (%)	Coal CV (%)	Interpretation
AND240-4253	34.2	47.0	Highly predictable
RN141-2106	68.0	86.0	Highly variable
MZA 1701	85.2	77.3	Variable
MZA 1801	87.6	71.4	Highly variable

Table 6: Coefficient of Variation (CV)

AND240-4253 is the most reliable locomotive for operational planning.

4.5 Comparative profiles

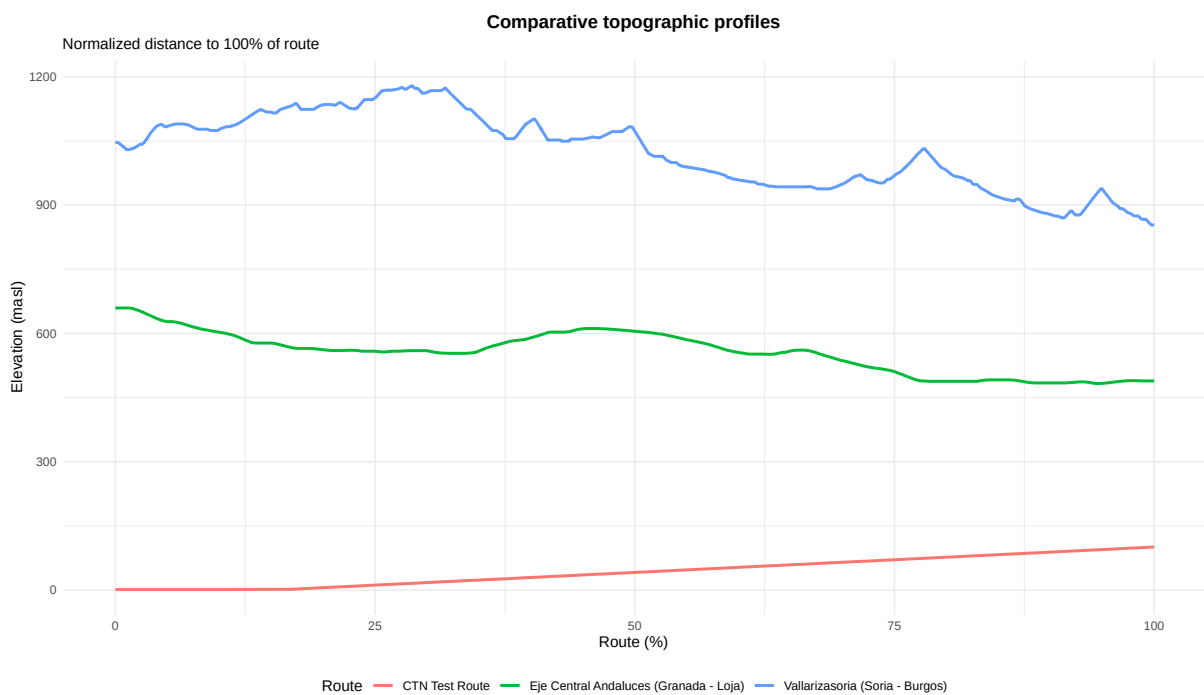


Figure 8: Comparative profiles

5. REGRESSION ANALYSIS

5.1 Coal vs Speed

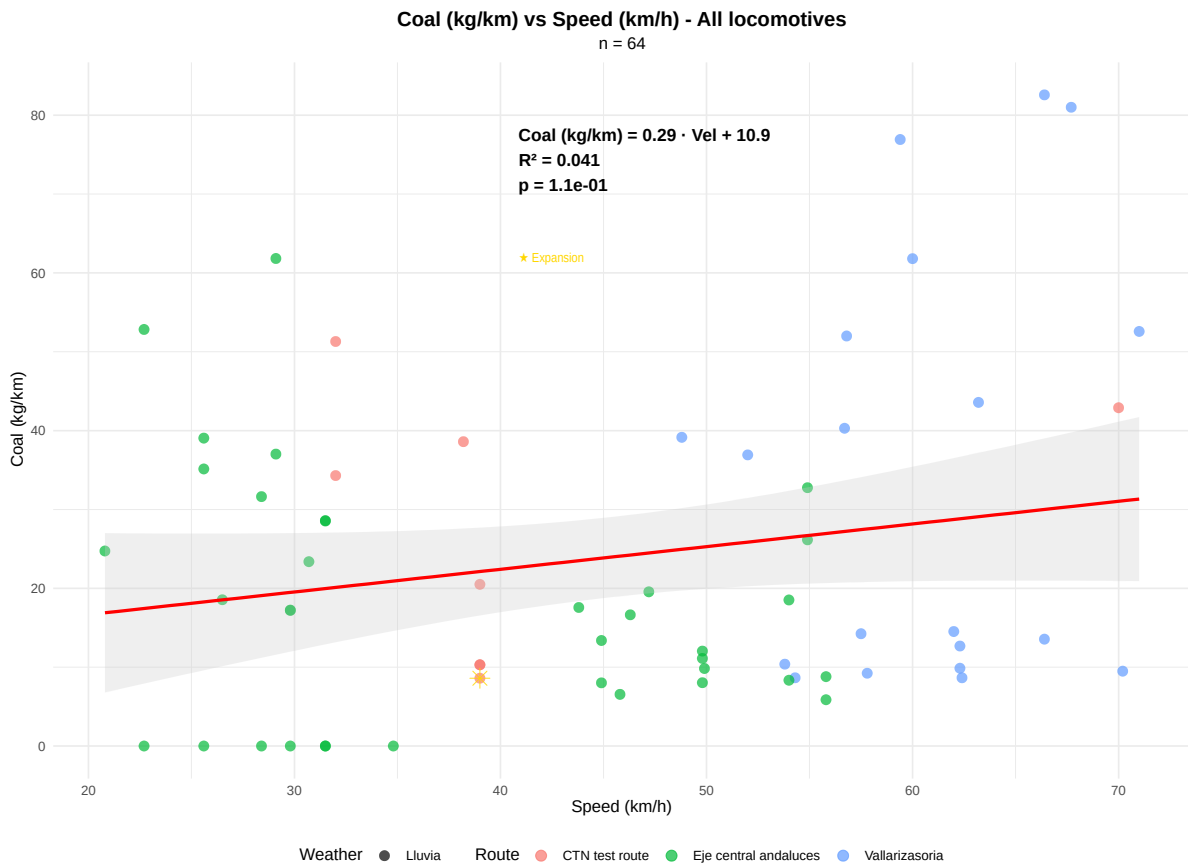


Figure 9: Coal vs Speed regression (all)

Locomotive	Slope (kg/km per km/h)	R²	Interpretation
MZA 1801	+0.775	0.218	Very sensitive
AND240-4253	+0.488	0.157	Sensitive
MZA 1701	-0.158	0.075	Stable (slight decrease)
RN141-2106	-0.019	0.020	Very stable

Table 7: Coal sensitivity to speed

MZA 1801 coal consumption increases sharply with speed. Mikado and MZA 1701 are virtually insensitive.

5.2 Water vs Speed

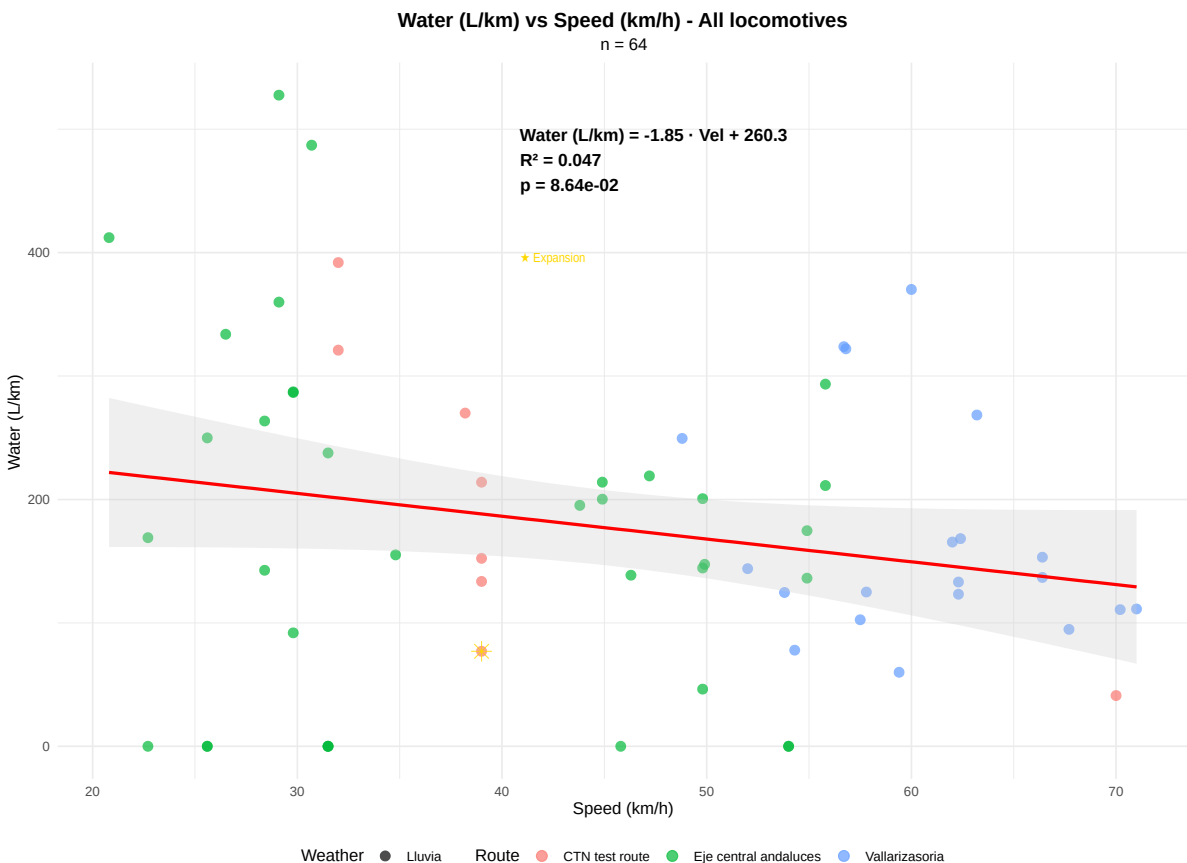


Figure 10: Water vs Speed regression (all)

Locomotive	Slope (L/km per km/h)	R ²	Interpretation
MZA 1801	-2.963	0.152	High saving at speed
AND240-4253	-1.796	0.054	Moderate saving
MZA 1701	-0.634	0.007	Slight saving
RN141-2106	-0.518	0.005	Very slight saving

Table 8: Water sensitivity to speed

All locomotives reduce water consumption at higher speeds (expansion driving effect). MZA 1801 benefits most.

6. CORRELATION: ZONE VS TRAMO

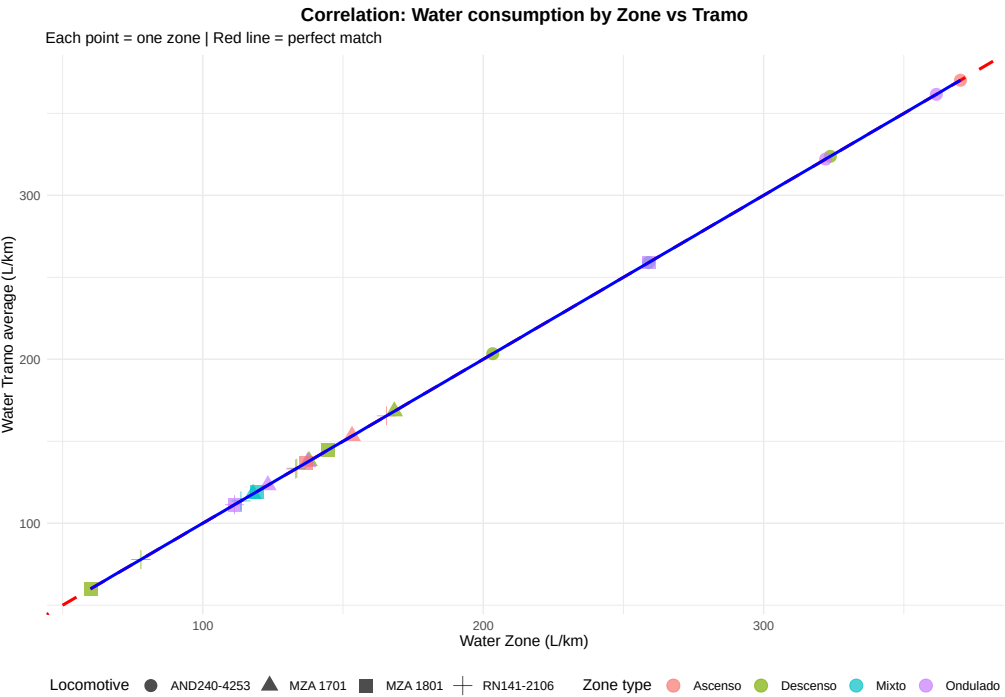


Figure 11: Water correlation zone vs tramo

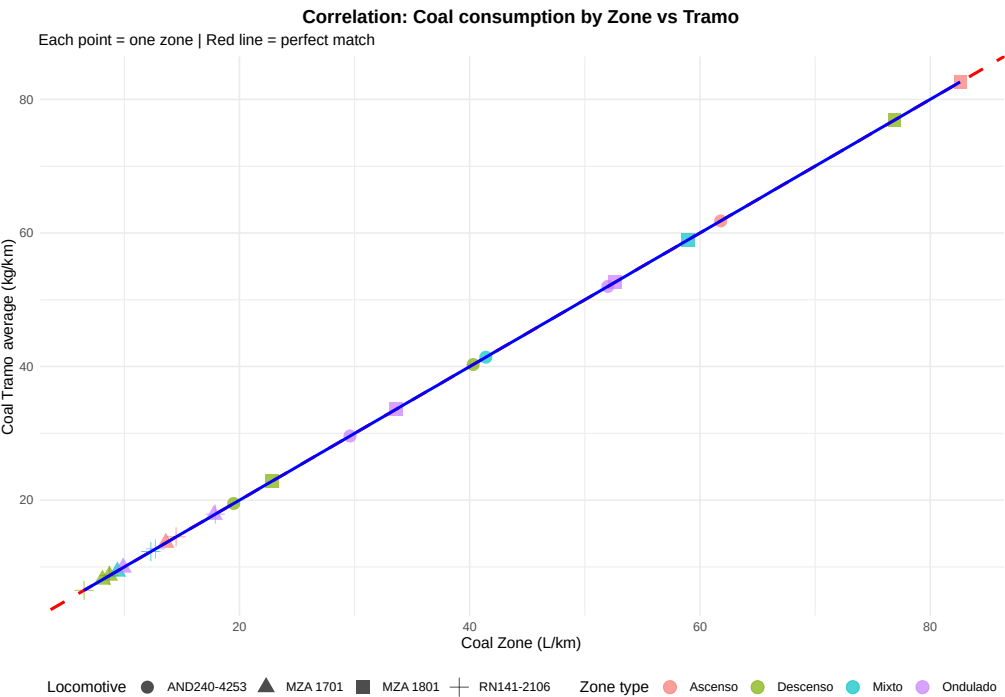


Figure 12: Coal correlation zone vs tramo

Parameter	Correlation (r)	Interpretation
Water	0.99	Almost perfect
Coal	0.98	Almost perfect

Table 9: Correlation between approaches

Zone-based and tramo-based analyses are virtually identical. Zones are a valid methodological tool.

7. CONCLUSIONS

7.1 Which locomotive to choose by priority?

If you prioritize ...	Choose ...	Key data
Efficiency (water+coal)	RN141-2106 (Mikado)	Most balanced
Economy (€/km)	MZA 1701	2.93 €/km for long distance
Speed	MZA 1801	47.8 km/h average
Predictability	AND240-4253	CV 34-47%, no surprises

Table 10: Recommendations by priority

7.2 Key findings

1. MZA 1801 consumes ~3x more coal than MZA 1701 (38.7 vs 12.3 kg/km)
2. AND240-4253 consumes ~2x more water than the Mikado (277.8 vs 133.9 L/km)
3. MZA 1801 is the fastest (+10% over the slowest)
4. AND240-4253 is the most predictable (CV 34-47%)
5. MZA 1701 is most efficient on climbs
6. The Mikado best utilizes inertia on descents

7.3 The AND240-4253 niche

Although the most inefficient locomotive in absolute consumption, its **predictability** (lowest CV) makes it valuable for:

- Short-distance services with frequent stops (commuter, freight)
- Refueling planning without excessive safety margins
- Scheduled maintenance without uncertainty

7.4 The MZA 1801 paradox

Designed as a high-speed locomotive (streamlined, 20 kg/cm² pressure), it operated at 90% pressure due to post-war conditions (low-quality oils). This explains its high coal consumption: **it wasn't inefficient by design, but due to the operational constraints of its era.**

8. OPERATIONAL RECOMMENDATIONS

Service	Recommended Locomotive	Reason
Long distance (e. g., Madrid-Barcelona)	MZA 1701 or Mikado	Lowest cost/km
Commuter/Regional (many stops)	AND240-4253	Predictable, easy to plan
Express services (speed priority)	MZA 1801	Fastest
Steep grades	MZA 1701	Best climb performance
No clear priority	RN141-2106 (Mikado)	Most balanced, works well everywhere.

Table 11: Recommendations by service type

9. APPENDICES

All materials are available in the repository:

- **R Scripts** (8 documented scripts)
- **Database** (SQL schema and anonymized data)
- **Graphics** (20 selected SVG figures)
- **Results tables** (CSV with statistics and analyses)
- **Historical validation** (driver testimony and technical article)

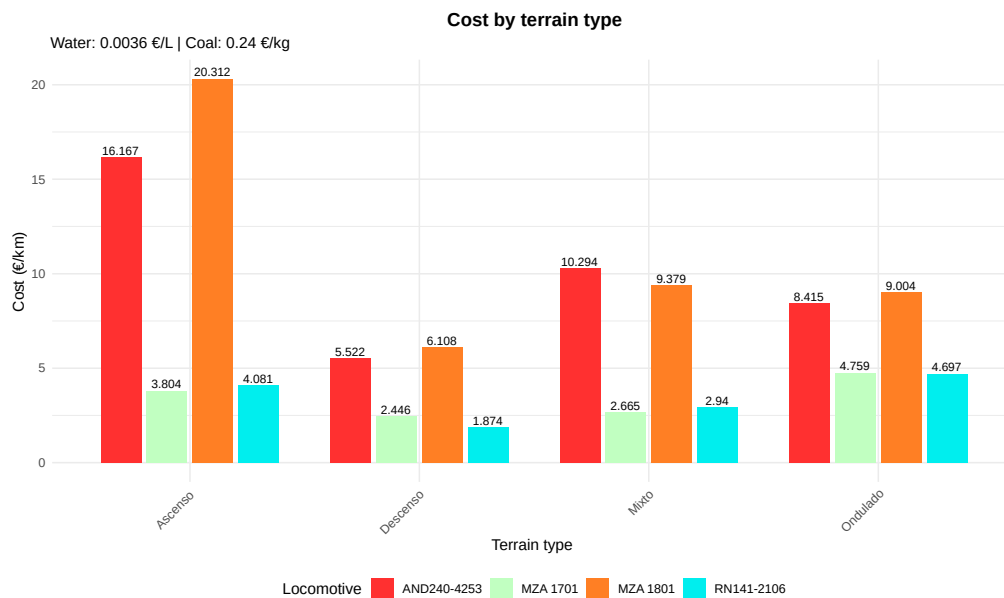


Figure 13: Cost by zone

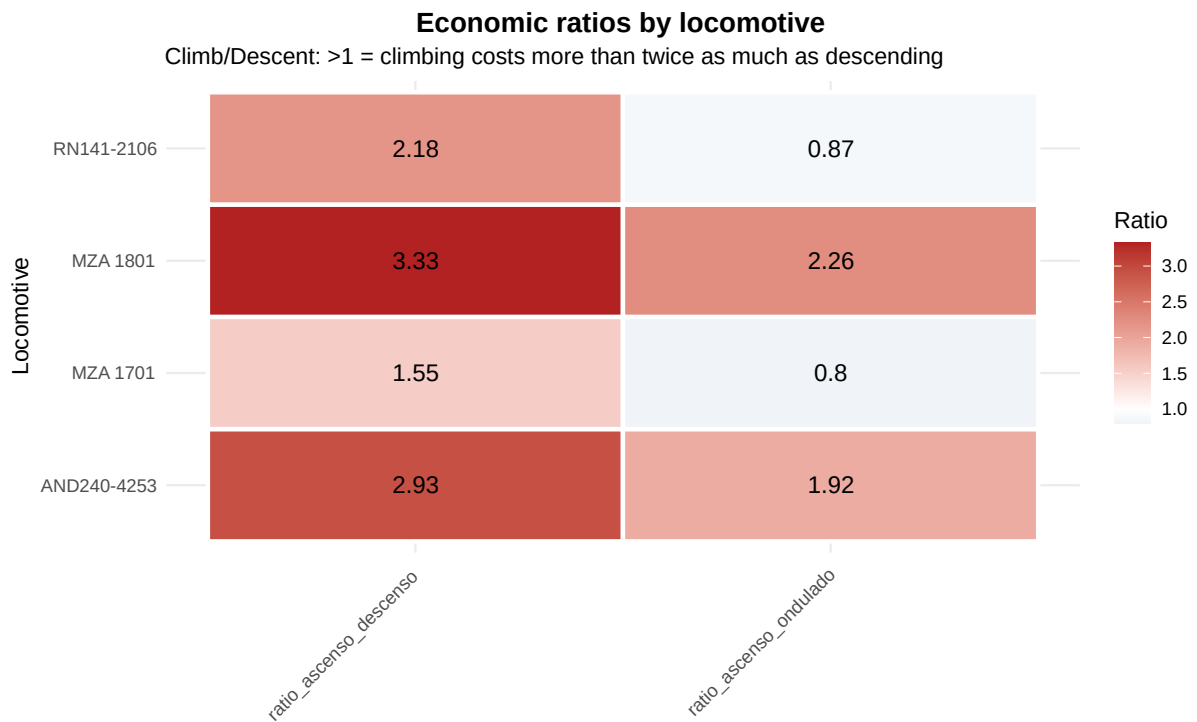


Figure 14: Economic ratios

10. FUTURE WORK (open research lines) 🌐

- Impact of coal quality (calorific value, ash content) on consumption
- Influence of water pH and hardness on thermal efficiency
- Wear and tear modeling over time
- Calibration of AND240-4253 as a test bench for experiments

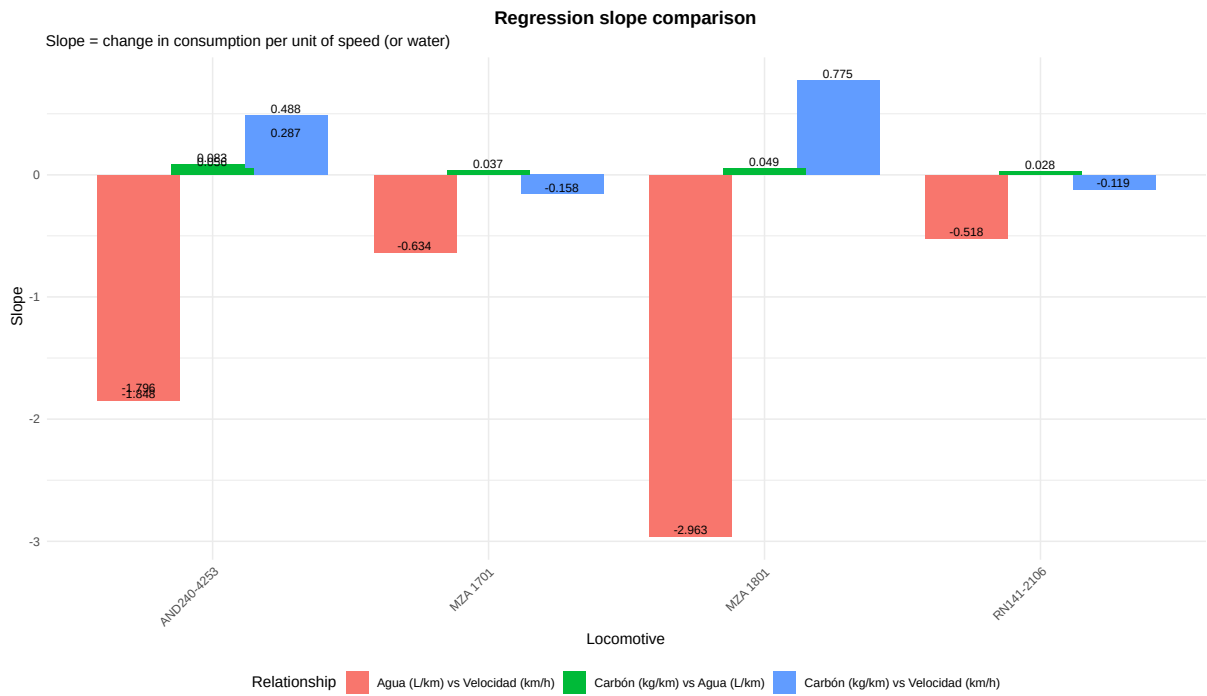


Figure 15: Slope comparison

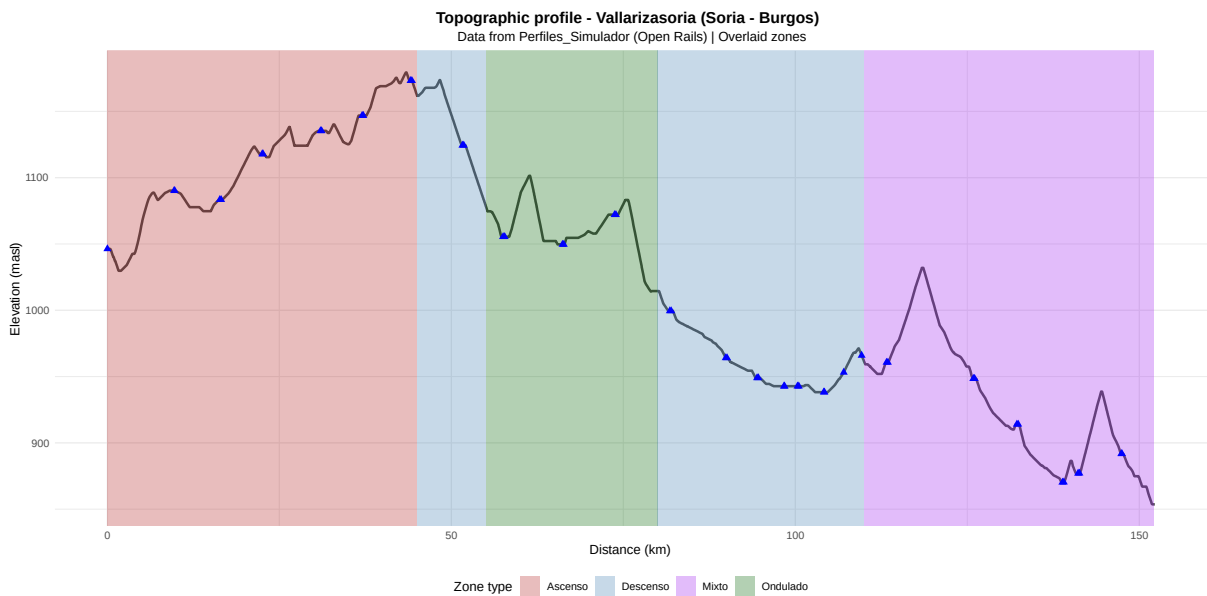


Figure 16: Vallarizatoria profile

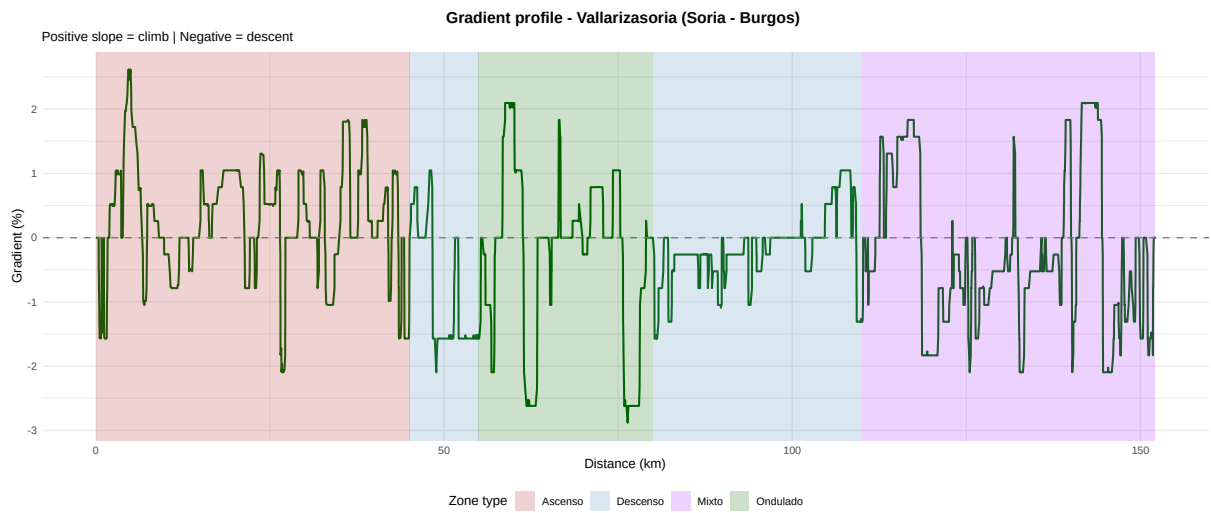


Figure 17: Vallarizatoria gradients